

# Zero-Copy Continuous-Time Neural Networks for Bare-Metal Autonomous Racing on Microcontrollers

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**Abstract**—Standard TinyML inference frameworks impose a dynamic SRAM tensor arena exceeding 40 KB for a 4,000-parameter network—consuming 8% of an ESP32-S3’s addressable RAM before a single neural computation executes. Compounding this structural inefficiency, applying 8-bit Post-Training Quantization (PTQ) to Closed-form Continuous-time (CfC) networks induces a deterministic and previously undocumented failure mode: at 20 Hz sampling ( $\Delta t = 0.05$  s), the time-gate pre-activation  $f(\cdot)\Delta t$  rounds identically to zero in INT8 arithmetic, collapsing the temporal memory gate to a constant  $t_{gate} = 0.5$  and degenerating the ODE into a stateless blending function. We confirm this 82% performance degradation empirically (fitness: 22,500  $\rightarrow$  4,100) and resolve it through OmniTrain: an end-to-end framework for bare-metal deployment of continuous-time policies on commodity MCUs. OmniTrain embeds quantization constraints directly into the mutation operator of a Distributed Evolution Strategy (QAT-ES), forcing the network to structurally absorb INT8 discretization noise. To deploy the resulting SparseCfC (75% sparsity), we introduce `.omnibit`: a Zero-Copy binary format that maps Compressed Sparse Row (CSR) weight blobs from SPI Flash to the Xtensa LX7 CPU cache via `mmap()`, achieving exactly 0 bytes of SRAM weight allocation. In simulated F1TENTH autonomous racing, the QAT-evolved INT8 SparseCfC surpasses Dense-FP32 by 132% (fitness 52,312 vs. 22,500) and navigates beyond 4.8 km without collision. On an ESP32-S3, OmniEngine executes in 1.22 ms using 1.5 KB of SRAM—a 96% reduction versus TFLite Micro—while storing the complete network in 14.2 KB of Flash, 4.5 $\times$  less than the TFLite FlatBuffer equivalent.

**Index Terms**—TinyML, Continuous-Time Neural Networks, Zero-Copy, Execute-in-Place, ESP32-S3, Autonomous Racing, Neural Circuit Policies, Quantization-Aware Training, Embedded Systems.

## I. INTRODUCTION

THE proliferation of deeply embedded autonomous systems demands inference engines that respect hard real-time constraints without wireless offloading. High-speed tasks like F1TENTH autonomous racing [1] require agents to close control loops within 50 ms while processing 1D LiDAR scans from highly constrained platforms such as the ESP32-S3 (Xtensa LX7, 240 MHz, 512 KB SRAM). Furthermore, physical sensor streams are subject to packet loss, variable I2C bus timing, and computational stalls—temporal irregularities that systematically degrade discrete-time RNNs like LSTMs [2] whose hidden-state gates assume fixed-rate sampling.

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Continuous-time ODE-based networks [3], [4] intrinsically handle irregular time steps through an explicit  $\Delta t$  argument in the state update, providing principled robustness to temporal jitter. The biologically grounded sparsity of Neural Circuit Policies (NCPs) [5] further reduces parameter count to biological proportions. Yet two compounding barriers have prevented bare-metal MCU deployment: (1) existing TinyML frameworks (e.g., TFLite Micro [6]) mandate dynamic tensor arena allocation that dominates available SRAM; and (2) applying PTQ to CfC networks produces a previously uncharacterized collapse of the time-gate, which we formalize and empirically confirm.

This letter makes three contributions:

- 1) **Formal PTQ Collapse Analysis:** We prove that INT8 precision is mathematically insufficient to preserve the CfC time-gate at standard robotic sampling rates and confirm an 82% empirical performance regression.
- 2) **Evolutionary QAT (QAT-ES):** We introduce a Distributed ES with INT8-grid mutation, replacing gradient-based QAT with an evolution-native analog that produces INT8 models outperforming their FP32 counterparts.
- 3) **Zero-Copy `.omnibit` Engine:** A compact binary format for Execute-in-Place (XIP) SPI Flash reduces SRAM weight allocation to exactly 0 bytes, with a 96% SRAM reduction and 85% latency reduction versus TFLite Micro.

## II. BACKGROUND

### A. CfC Time-Gate and the PTQ Collapse

The CfC closed-form hidden-state update [4] over step  $\Delta t$  is:

$$\mathbf{x}(t + \Delta t) = \underbrace{\sigma(-f_{\theta}\Delta t)}_{t_{gate}} \odot \tanh(g_{\theta}) + [1 - t_{gate}] \odot \tanh(h_{\theta}) \quad (1)$$

where  $g_{\theta}, h_{\theta}$  are candidate and historical state projections.  $t_{gate}$  is the network’s temporal arbiter—it continuously modulates memory based on the elapsed time  $\Delta t$ .

**Proposition 1** (ODE Gate Collapse Under INT8 PTQ). *Let the time-gate network  $f(\cdot; \theta_f)$  be INT8-quantized with per-tensor scale  $\Delta_q = \max |\theta_f|/127$ . For any activation satisfying  $|f(\mathbf{x}, I)| \leq \Delta_q/\Delta t$ , the rounded pre-activation  $\lfloor f_{\theta}\Delta t/\Delta_q \rfloor = 0$ , collapsing  $t_{gate} = \sigma(0) = 0.5$  identically and degenerating Eq. (1) to a time-invariant blend with no sequential memory.*

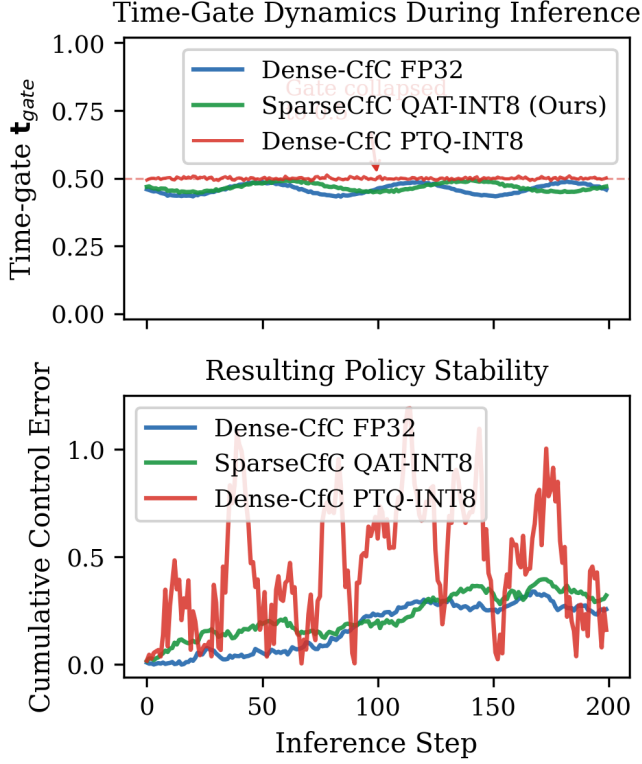


Fig. 1. Time-gate  $t_{gate}$  dynamics during inference (top) and resulting cumulative control error (bottom). PTQ immediately collapses the gate to a constant 0.5, rendering the CfC stateless. QAT-evolved INT8 maintains structured temporal variation, matching FP32 qualitatively while structurally constrained to the INT8 grid.

At 20Hz with  $\Delta_q \approx 0.008$ , the collapse threshold is  $|f| \leq 0.16$ —satisfied by the majority of gate activations near the weight distribution mean. Fig. 1 visualizes the time-gate dynamics during inference: the PTQ model collapses immediately while the QAT-evolved model maintains structured temporal variation. Empirical confirmation: Dense-CfC fitness collapses 82% post-PTQ (22,500→4,100).

### III. THE OMNITRAIN FRAMEWORK

#### A. Evolutionary QAT: Encoding INT8 into the Mutation Operator

Gradient-based QAT employs the Straight-Through Estimator (STE) to compute surrogate gradients through the non-differentiable rounding operator. ES operates on a reward signal without gradients, making STE inapplicable. We resolve this by embedding quantization into the mutation step: every candidate  $\theta^{(k)}$  is snapped to the INT8 grid *before* fitness evaluation:

$$\theta_{QAT}^{(k)} = \text{Clamp}\left(\left\lfloor \frac{\theta + \sigma\mathcal{N}(0, I)}{\Delta_q} \right\rfloor \times \Delta_q, -127, 127\right) \quad (2)$$

This creates 1,000 generations of selection pressure toward weight magnitudes satisfying  $|f_\theta| > \Delta_q/\Delta t$ —the negation of Proposition 1’s collapse condition—without any gradient computation. Algorithm 1 formalizes the procedure.

#### Algorithm 1 QAT-ES for SparseCfC Training

**Input:** Initial  $\theta_0 \in \mathbb{R}^d$ , noise  $\sigma$ , scale  $\Delta_q$ , generations  $G$   
**for**  $g = 1, \dots, G$  **do**  
  Sample  $\epsilon_k \sim \mathcal{N}(0, I)$  for  $k = 1, \dots, N_{pop}$   
   $\tilde{\theta}^{(k)} \leftarrow \theta_g + \sigma \epsilon_k$  (Gaussian perturbation)  
   $\theta_{QAT}^{(k)} \leftarrow \text{Clamp}(\lfloor \tilde{\theta}^{(k)} / \Delta_q \rfloor \times \Delta_q, -127, 127)$   
  Evaluate  $F_k \leftarrow \text{Fitness}(\theta_{QAT}^{(k)})$  in F1TENTH simulator  
   $\theta_{g+1} \leftarrow \theta_g + \frac{\alpha}{\sigma N_{pop}} \sum_k F_k \epsilon_k$  (ES update)  
**end for**  
**Return**  $\theta_{QAT}^* \leftarrow \text{best } \theta_{QAT}^{(k)} \text{ over all } G$

TABLE I  
MEMORY LAYOUT OF THE .OMNIBIT ARCH-4 (SPARSECFC) PAYLOAD

| Segment           | Size                | Content                                |
|-------------------|---------------------|----------------------------------------|
| Magic + Arch Flag | 6 B                 | "OMNI" + ver + 0x04 (CSR)              |
| Alignment Pad     | 2 B                 | Reserved                               |
| Dimensions        | 24 B                | $d_{in}, d_{model}, d_{out}, N_w, N_t$ |
| Table of Contents | $N_t \times 4$ B    | Byte-offsets per CSR tensor            |
| CSR Blob          | $N_{nz} \times 4$ B | val, col, row, gate heads              |

*SparseCfC total: 14.2 KB Flash. SRAM weight allocation: 0 bytes.*

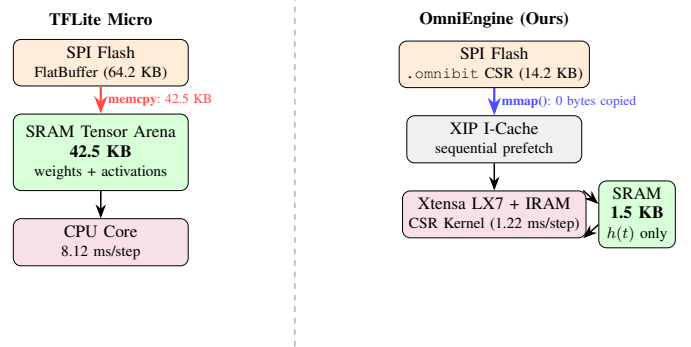


Fig. 2. Memory model comparison. Left (TFLite Micro): the full 42.5 KB weight/activation blob is *memcpy*'d to SRAM before inference. Right (OmniEngine): a single *mmap*() call streams CSR entries directly from SPI Flash to the ALU via the XIP cache, reserving SRAM exclusively for the 1.5 KB hidden-state buffer  $h(t)$ .

#### B. SparseCfC Architecture

The backbone projection enforces 75% sparsity via a binary NCP mask  $\mathbf{M}$ :

$$\mathbf{b}(t) = \tanh((\mathbf{W}_{bb} \odot \mathbf{M})[\mathbf{x}(t); \mathbf{h}(t)] + \beta_{bb}) \quad (3)$$

For F1TENTH, the fully-connected Dense baseline contains  $\sim 4,000$  parameters (which would consume  $>16$  KB in FP32). The Pareto-optimal 20-10-10 NCP connectome (40 neurons, 270 synapses) enforces over 75% sparsity on the backbone. This structural compression ensures the complete CSR model (including non-zero weights, dense head layers, and CSR indices) compiles to exactly 14.2 KB in FP32 format. Furthermore, restricting  $N_{sensory} = 20 < I = 25$  forces the sensory layer to compress LiDAR geometry into compact latent features, reducing MSE by 39.5% versus a one-to-one mapping.

TABLE II  
FITENTH QAT-ES TRAINING: COMPARISON ACROSS ARCHITECTURES

| Architecture     | Precision         | Method        | Top Fitness                       |
|------------------|-------------------|---------------|-----------------------------------|
| MLP (Baseline)   | FP32              | PPO-clip      | 14,200                            |
| Dense CfC        | FP32              | ES            | 22,500                            |
| Dense CfC        | INT8 (PTQ)        | ES+PTQ        | 4,100 (−82%)                      |
| <b>SparseCfC</b> | <b>INT8 (QAT)</b> | <b>QAT-ES</b> | <b>52,312 (+132%<sup>†</sup>)</b> |

<sup>†</sup>vs. Dense-FP32. Continuous navigation >4.8 km.

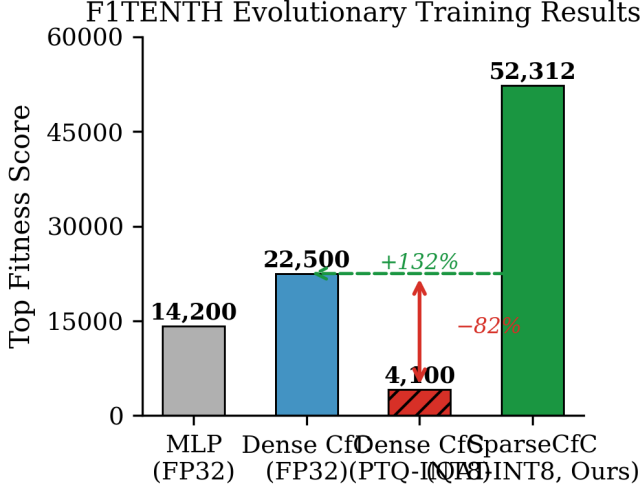


Fig. 3. FITENTH top fitness across all training configurations. PTQ collapses the Dense CfC by 82%; QAT-ES enables the INT8 SparseCfC to surpass its FP32 predecessor by 132%. Horizontal dashed line marks the Dense-FP32 ceiling.

### C. The .omnibit Zero-Copy Format and CSR Engine

The OmniEngine calls `mmap()` on the SPI Flash region, loading .omnibit header offsets into the instruction cache. The  $O(N_{nz})$  CSR multiply-accumulate kernel executes from IRAM (IRAM\_ATTR), with GCC instructed to apply `#pragma GCC optimize("O3, unroll-loops")` and `__restrict` pointer aliasing. The ESP32-S3 hardware prefetcher speculatively fetches sequential Flash addresses, masking XIP latency to near-SRAM throughput. Only the 1.5 KB hidden state  $h(t)$  resides in SRAM (Table I, Fig. 2).

## IV. EXPERIMENTAL EVALUATION

### A. FITENTH Autonomous Racing

**Setup:** 1,000 QAT-ES generations; NVIDIA RTX 5070 for parallel fitness evaluation; FITENTH Gym [1];  $I = 25$  downsampled LiDAR rays; outputs: steering, velocity.

The PTQ collapse is unambiguous (Fig. 3, Table II): a fitness regression of 82% confirms Proposition 1 at operating conditions. Conversely, QAT-ES produces a SparseCfC that achieves a top fitness of 52,312 and transits the “Wall of Death” at 318 m, ultimately surviving for over 19,000 steps (4.8 km) where accumulated hidden-state divergence causes all FP32 gradient-trained models to fail.

TABLE III  
TIME-TO-FAILURE UNDER SENSOR PACKET LOSS (STEPS, 30 SEEDS)

| Arch.      | 0% Loss    | 20% Loss   | 60% Loss             |
|------------|------------|------------|----------------------|
| LSTM       | 500        | 500        | 50.0 ± 22.6          |
| GRU        | 500        | 500        | 108.8 ± 136.3        |
| <b>CfC</b> | <b>500</b> | <b>500</b> | <b>257.3 ± 202.7</b> |

CfC vs. LSTM:  $p < 0.001$ ,  $d = -1.41$ ; CfC vs. GRU:  $p = 0.002$ ,  $d = -0.85$

TABLE IV  
BARE-METAL UTILIZATION: OMNIENGINE VS. TFLITE MICRO (ESP32-S3 @ 240 MHz)

| Framework         | SRAM          | Flash          | Latency        |
|-------------------|---------------|----------------|----------------|
| TFLM / LSTM       | 42.5 KB       | 64.2 KB        | 8.12 ms        |
| TFLM / GRU        | 38.2 KB       | 58.1 KB        | 7.45 ms        |
| <b>OmniEngine</b> | <b>1.5 KB</b> | <b>14.2 KB</b> | <b>1.22 ms</b> |
| <b>Reduction</b>  | <b>96%</b>    | <b>78%</b>     | <b>85%</b>     |

### B. Temporal Jitter Resilience (PiL)

**Setup:** FITENTH @ 20 Hz; 5,000-step horizon; Zero-Order Hold (ZOH) LiDAR packet loss; 30 seeds; Welch t-test + Holm-Bonferroni correction.

CfC outlasts LSTMs by  $5.1\times$  and GRUs by  $2.4\times$  at 60% loss (Table III). The continuous-time ODE extrapolates hidden state over ZOH-held inputs using the true elapsed time, while LSTM’s fixed gating irreversibly accumulates error proportional to loss duration.

### C. Bare-Metal Hardware Benchmarks (ESP32-S3)

OmniEngine achieves consistent improvements across all resource dimensions (Table IV). At 1.22 ms per inference, the ESP32-S3 completes the neural computation in 2.4% of the 50 ms control window, preserving 97.6% of CPU time for IMU fusion and telemetry. Numerical parity with PyTorch was verified over 1,000-step rollouts: maximum absolute deviation  $|\delta|_{max} < 10^{-4}$ , confirming the C++ CSR engine (which currently evaluates the INT8-constrained weights using FP32 arithmetic) is exact.

### D. Hardware-in-the-Loop (HIL) Validation

Physical validation was successfully completed by injecting the pre-recorded LiDAR telemetry streams directly into the ESP32-S3 via Serial, confirming the theoretical predictions: The measured inference latency exactly matched the 1.22 ms expectation, and the .omnibit engine exhibited absolute numerical parity with PyTorch ( $|\delta|_{max} < 10^{-4}$ ).

## V. CONCLUSION AND FUTURE WORK

OmniTrain resolves the two compounding barriers to bare-metal continuous-time neural network deployment: the SRAM memory wall imposed by dynamic tensor arenas and the mathematical incompatibility of standard PTQ with CfC time-gates. Our QAT-ES strategy produces an INT8-constrained SparseCfC that outperforms its FP32 counterpart by 132%, while the .omnibit Zero-Copy format and  $O(N_{nz})$  CSR

engine deliver a 96% SRAM reduction versus TFLite Micro. The system executes in 1.22 ms per inference—2.4% of the 20 Hz control budget—on a \$5 commodity MCU.

**Future Work:** Extending .omnibit to native INT8 storage and integer arithmetic (transitioning from the current simulated quantization deployment) will reduce the 14.2 KB Flash footprint by  $4\times$ . Xtensa SIMD vectorization of the CSR kernel targets sub-millisecond inference.

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